

Muhammad Waqas

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Research Profile

A computational researcher with a Ph.D. in Computer Science specializing in simulation and analytical modelling of complex systems for optimization. My doctoral research focused on developing sophisticated simulation models to analyze resource allocation and efficiency trade-offs in large-scale, capacity-constrained computing systems. This work built upon my foundational experience in network science, where I applied social network metrics to successfully optimize routing protocols. This dual expertise in system simulation and network analysis drives my passion for interdisciplinary research. I am now eager to apply my quantitative skills to pressing challenges at the intersection of technology and society

Education

- 2025 **Ph.D. in Computer Science**, *Ca' Foscari University of Venice*, Venice, Italy
Thesis: "Optimizing Cloud Computing Performance: Simulation of Finite-Capacity Multi-Server Queueing Systems"
 - Designed and implemented an open-source simulation framework to model finite-capacity systems
 - Developed and validated novel scheduling algorithms (preemption strategies), achieving a 30% reduction in mean wait times while analyzing the trade-offs between QoS and system efficiency
 - Conducted rigorous statistical analysis of simulation outputs to quantify the impact of different scheduling policies on overall system behavior and resource utilization
- 2017 **Masters in Software Engineering**, *COMSATS University Islamabad*, Islamabad, Pakistan
Thesis: "Analysis of Convergence Time in OSPF Routing Protocol Using Social Network Metrics"
 - Applied principles from Social Network Analysis, utilizing graph theory and centrality metrics to identify critical nodes in network topologies.
 - Developed a Java-based "Topology Analyzer" tool to implement this interdisciplinary approach.
 - Demonstrated a 22% reduction in OSPF convergence time as a result of network optimizations

Job Experience

- 2019 – 2021 **Lecturer**, *University of Lahore*, Islamabad, Pakistan
Taught undergraduate courses, supervised projects, and contributed to NCEAC accreditation and curriculum planning
 - Courses: Software Engineering, HCI, Requirements Engineering, Discrete Structures, ICT
 - Admin Roles: Batch Advisor, Member – Course Folder Review Committee
- 2017 – 2019 **Lecturer**, *IQRA National University*, Peshawar, Pakistan
Delivered courses, supervised projects, and participated in academic committees
 - Courses: Formal Methods, Software Engineering, HCI, Enterprise Systems
 - Admin Roles: Member – Board of Studies, revised 3 course syllabi

Publications

Peer Reviewed

- [1] S. Akbar, R. Li, **M. Waqas**, and A. Jan, "Server temperature prediction using deep neural networks to assist thermal-aware scheduling," *Sustainable Computing: Informatics and Systems*, Elsevier, vol. 36, art. no. 100809, 2022
 - [2] **M. Waqas**, S. UrRehman, and S. Akbar, "Convergence time analysis of OSPF routing protocol using social network metrics," *Future Generation Computer Systems*, Elsevier, vol. 94, pp. 62-71, 2019
- Conferences
- [1] **M. Waqas**, A. Marin, and L. Maccari, "Finite capacity multi-server job systems: A simulation study" in *Proc. 38th Eur. Conf. on Modelling and Simulation (ECMS)*, Krakow, Poland, June 2024, pp. 150-157
- Submitted
- [3] **M. Waqas**, A. Marin, L. Maccari, and M. Raza, "Simulating Finite-Capacity Multiserver Queues in OMNeT++" submitted to *Future Generation Computer Systems*, Elsevier, 2025
 - [4] **M. Waqas**, A. Marin, and L. Maccari, "When killing jobs becomes good?" submitted to *Journal of Parallel and Distributed Computing*, Elsevier, 2025

Additional Training

- Coursera **Introduction to Data Analytics** (IBM)
- Coursera **Introduction to Cloud Computing** (IBM)
- HEC **Outcome based Education** (NCEAC)
- UniVe **Intellectual Property and Technology Transfer** (PInK)
- UniVe **ZOTERO: Advanced organization and management of bibliographies**

Technical Skills

- Network Science NetworkX, Pajek, Social Network Analysis, Graph Theory
- Programming & Analysis Python, MATLAB, Java, C++
- Simulation & Modeling OMNeT++, NS-3, CloudSim, Formal Methods (Model Checking, Z notation)
- Tools & Platforms Git, VS Code, L^AT_EX, CZT Toolkit

References

- Dr. Andrea Marin Full Professor, Department of Environmental Sciences, Informatics and Statistics
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- Dr. Saif ur Rehman Malik Assistant Professor, School of Computer Science and Statistics
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